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(1) Research Program/Project/Study Title: Shellwaste Management of Fisheries Sector in Roxas City: Input to Circular Economy	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center						
(4) Site of implementation <table border="1" style="width: 100%;"> <tr> <td>Municipality/City:</td> <td>Roxas City</td> </tr> <tr> <td>Province:</td> <td>Capiz</td> </tr> <tr> <td>Region:</td> <td>VI Western Visayas</td> </tr> </table>			Municipality/City:	Roxas City	Province:	Capiz	Region:	VI Western Visayas
Municipality/City:	Roxas City							
Province:	Capiz							
Region:	VI Western Visayas							
(5) Classification <table border="1" style="width: 100%;"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:							
☒ Basic	☐ Pilot Testing							
☐ Applied	☐ Tech. Promotion/Commercialization							

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(7) Sector/Commodity Environment/Fisheries	(8) Discipline/Field of Specialization Environmental Science
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) Roxas City is known as the country's "seafood capital" due to its abundant fish and other seafood production encompasses of freshwater and saltwater fish, molluscan shellfish, cephalopod mollusks and crustaceans . The variety of shellfishes in the locality became an attraction to most tourists. Fishing communities in the city value these seashells for meat as consumption. However, empty seashells found no value and discarded as wastes in landfill or the sea . Thus, contributing to solid waste problems in the community. Contamination of the ecosystem by bivalve shells is one of the major problems with which the world's largest producers have already met, and which could endanger further sustainable development of this branch of industry. Enormous amount of waste seashells were dumped into public waters and/or landfills, and cause many environmental problems including pollution of coastal fisheries, management of public water surface, a bad smell as a consequence of the decomposition of organics attached to the shells, damage of natural landscape and health/sanitation problems (Hou et al., 2016). As the production and processing of bivalves have increased, efficient use of their shells has become essential, not only to maximize financial return, but also to address waste disposal problems because of their slow natural degradation rate. The most heavily traded bivalve mollusk species are mussels, clams, scallops and oysters, mostly farmed, and world bivalve production (both wild and farmed) of selected species in 2016 were: oysters 54%, scallops 24%, mussels 18% and calms 4% (GLOBEFISH, 2018). A possible problem must be considered to prevent potential environmental damage and also to avoid additional costs associated with the rehabilitation and treatment of such waste. Finding solutions to the accumulation of seashell waste would reduce ecological and financial problems. Circular economy is based on the idea that there is no such thing as waste wherein products are designed for another potential useful usage (Shatel, 2016). This study aims to explore the shell waste management practices of the fisheries sector in Roxas City and offer new perspectives and direction on value adding waste shell-derived.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)	

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In order to explore the shellwaste management of the fisheries sector in Roxas City the following activities will be employed.

1. Identification of sector/industry utilizing shell fishes.
2. Field observation
3. Interview
4. Focus-discussion
5. Development of waste management and utilization plan

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Generally, this study aims to explore the sea shell waste management practices of the fisheries sector in Roxas City and offer new perspectives and direction on value adding waste shell-derived. Specifically, this study aims to:

1. Determine the sea shell waste generation, disposal, management and practices of the fisheries sector of Roxas City.
2. Identify issues and risks from seashell waste.
3. Ascertain ways for reducing seashell waste accumulation.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Bivalve mollusk farming now makes up a significant portion of the worldwide aquaculture business, accounting for 23% of global production and showing signs of continuous expansion. This production is a globally interesting source of affordable, nutritious food for the planet's expanding human population. However, as bivalve shells can make up as much as 75% of the total weight of a bivalve, one of the main issues facing this sector of the economy is the contamination of the ecosystem by bivalve shells. Large amounts of seashell garbage are created all around the world and frequently abandoned in landfills or the ocean. This garbage accumulates in coastal areas and contributes to numerous environmental issues and even in health, such as the occurrences of disease and noxious odor particularly concerning in the cases of illegal dumping in various places and open storage of waste shells (Chilakala, et al, 2019). Thus, sea shell waste also needs better management.

Many fishing communities, households, shell-stall-owners in the market, local seafood restaurants in the Province of Capiz do not value empty seashells from oyster shells. After eating and/or selling its meat, empty shells are thrown away and are thought to be useless. Waste from empty shells of this mollusks derive from these major sources together with other solid waste like discarded glass and empty plastic bottles generated by the major business establishments, households, fast food chains and local seafood restaurants within the province have greatly contributed to the tons of solid wastes generated each year. This becomes a serious problem on Solid Waste

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Management by the Local Government Unit (LGU) and in the entire province (Hapinat, et al, 2019).

Shells are produced by molluscs as an outer skeleton, largely composed of a limy material, calcium carbonate. Molluscs are soft-bodied invertebrate animals which secrete this shell-building material. There are three major classes of molluscs well-known to many people: the univalves (Gastropoda) which include the snails, conches and periwinkles; the cephalopods (Cephalopoda), the squids, cuttlefishes, nautilus and octopuses; and the bivalves (Bivalvia), the clams, mussels, oysters and scallops. The shells produced by representatives of all three groups are potential raw materials for local trade. Various items such as buttons, necklaces, lampshades, flowers and other ornamental objects are made from the inedible shells. Shells which are not suitable for ornamental objects due to their poor quality find their way into lime making (Espejo-Hermes, 1985).

The discarded empty shells of oysters, Pecten and Mangrove clam shells contain potential compound known as Calcium Carbonate (CaCO_3). The high content of calcium carbonate in oyster shells, pecten and mangrove clam shells is technically possible for lime production to replace the commercial (CaCO_3) (Hamester, et al. 2012). Shells are a calcium rich source that produce Calcium Oxide (CaO) or lime, which is a major panacea in the problem of high soil acidity. They can supplant limestone as a source of commercial agricultural lime. Unlike stone lime which has a long-term impact on the deterioration of physical surroundings due to quarrying, shell lime is renewable and eco-friendly. It has proven uses in the environment such as increasing soil pH, improved soil nutritional status metrics including available phosphate and organic matter mass. Powdered shells are effective liming materials for crop cultivation and in eradication of giant earthworms infecting rice farms.

Also, in many coastal settlements around the globe, shells (most often and traditionally oyster shells) have been used in construction and many such buildings today represent popular tourist attractions. Powdered shells have been used for medicinal purposes and have been historically attributed with a myriad of health benefits and healing powers.

Reusing and recycling of such waste can be an excellent alternative to disposal. The seashell material attracts attention due to high calcium carbonate content, low-cost and availability provided by the fast developing seafood industry (Barros et al., 2009).

Alternative materials, such as seashell waste, are increasingly considered to reduce the exploitation of the reserves of natural limestone - one of the most exploited resources on the planet. In that sense, the utilization of seashell waste is a useful strategy for both sustainable resource management and reduced waste storage. Seashells contribute to more than 7 million tons of "nuisance waste" discarded every year by the seafood industry that mostly winds up thrown into landfills or dumped into

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the ocean. Reusing and recycling of such waste can be an excellent alternative to disposal. The seashell material attracts attention due to high calcium carbonate content, low-cost and availability provided by the fast developing seafood industry (Barros et al., 2009). Over the past decades, numerous articles have been published on the subject of shell valorization, citing a variety of potential applications that could alleviate the burden of waste shells on aquaculture and food producers, and in some cases present economic as well as environmental incentives to do so (Morris et al., 2019).

Several large-scale shell valorization strategies are currently exploited, and most of them are established in areas that generate large amounts of shell waste, i.e., where mutually beneficial partnerships have been established between shell producers and other industries (Morris et al., 2019). Further, some products from sea shells are already commercially available for a variety of applications.

Galicia (Spain) is the leading region in Europe that currently utilize seashell waste as a liming agent, because of the proximity of agricultural land to large shellfish aquaculture sites and the seashell processing facility (Barros et al., 2009; Morris et al., 2019). Until now, the uses of seashells have been limited to several established and sustainable applications (soil conditioners, low-cost adsorbents, calcium supplements, construction materials, etc.).

Also, numerous utilization possibilities were either discussed in the literature or have already been tested under realistic conditions but without commercialization (catalysts for biodiesel, bactericidal agents, artificial bones, dehalogenating agents, etc.). Today, the idea of seashell returning to the marine environment for conservation reasons became more and more popular and accepted among different organizations and research groups. A long-term effort is needed to seek possible recycling methods for shell waste and eliminate the environmental problems it can cause (Yao et al., 2014).

Historically, shells have been an important part of human culture. There is a variety of examples of mollusk shell use across the globe throughout human history, even 100,000 years ago. In prehistoric Stone Age sites in China and throughout Africa, seashells were used as a form of currency. Also, in many coastal settlements around the globe, shells (most often and traditionally oyster shells) have been used in construction and many such buildings today represent popular tourist attractions. Powdered shells have been used for medicinal purposes and have been historically attributed with a myriad of health benefits and healing powers. In African countries, wearing a shell amulet was thought to help maintain health, fertility, and luck. Shell amulets coated in gold and silver found in Egypt, and worn in recent history, was also intended to promote good health. The oldest identified piece of jewelry in the world was discovered in Israel and was made from the shells of the sea snail. There are many records on the use of seashells as tools for a variety of purposes including hammering, chipping, chopping, cutting, bevelling, etc. Furthermore, the shells of sea snail were used by native Australians as devices for expelling water from the canoes and as cooking pots (Morris, 2017).

Seashells can be exploited in numerous ways. Valorization is the principle of assigning value or giving greater value to something, where value can be seen from

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an economic, social and/or environmental perspective (Morris, 2017). Such concept is particularly relevant with the recent drive towards recycling, zero waste industries, and a circular economic system (European Commision, 2015).

The most common management practice to ameliorate acid soils is the surface application of lime or other calcareous materials (Bolan et al., 2003; Yao et al., 2014). As the main aim of soil liming is to neutralize acidic inputs, shell waste can be considered as an adequate substitute for this process. Seashell has been used as a liming agent alone (Lee et al., 2008), combined with other amendments to improve soil fertility (Kwon et al., 2009; Paz-Ferreiro et al., 2012), or used to remediate heavy metals pollution in soils (Ok et al., 2011). Teixeira et al. (1997) used mussel shells (*Mytilus galloprovincialis*) as a liming agent in soils in Galicia (Spain) and found that 9 t/ha of mussel shell had a comparable short-term positive effect on soil acidity as conventionally used magnesium limestone, but in the longer term, they proved to be less effective than mined liming agents in terms of soil fertility. Application of seashells to acidic soil resulted in the increased pH, soil organic matter, available P, exchangeable cations concentrations, and improved soil chemical and biological properties and increased plant productivity (Lee et al., 2008). Álvarez et al. (2012) reported that application of mussel shells to acidic soil increased the soil pH and content of exchangeable Ca, decreased the saturation of Al in the exchange complex, and had a positive effect on dry matter yield and Caconcentration in the plants. Addition of seashell waste was found effective in reducing Cu concentrations available to plants in Cupolluted vineyard soil (Fernández-Calviño et al., 2017), decreasing bioavailability of Pb in an army firing range soil (Ahmad et al., 2014), and increasing Cu, Cd, Ni and Zn retention in copper mine soil (Ramírez-Pérez et al., 2013). Ramírez-Pérez et al. (2013) demonstrated that the addition of the 24 g/kg mussel shells could retain Cu, Cd, Ni and Zn within the first few centimeters of the soil column length, indicating the usefulness of ground mussel shells to decrease the mobility and availability of these pollutants drastically. The results obtained in this study showed that mussel shells facilitate remediation efforts and further restoration of contaminated acidic soils. Shen et al. (2018) demonstrated that oyster shell powder application was more effective at reducing the incidence of tobacco bacterial wilt and improving the soil pH than lime and biochar. These authors recommend oyster shell powder as a soil amendment in order to improve soil pH and increase the bacterial richness and diversity of acidic tobacco-growing soils and thus contribute to the suppression of tobacco bacterial wilt. Paz-Ferreiro et al. (2012) demonstrated that mussel shell added to Cambisol significantly improved soil pH, lowered the amount of Al held in the cation exchange complex, increased soil microbial activity and enhanced soil fertility. However, some previous studies have shown certain limitations to the use of seashell as a liming amendment, such as a possible increase of Na content held in the cation exchange complex (Lee et al., 2008).

Oils and fats not suitable for human consumption can be used for biodiesel production. Unfortunately, such materials usually cannot be used directly as fuel due to their high viscosity, unsuitable combustion, poor atomization and can cause severe functional problems in an engine (Hou et al., 2016). The molecular weight of the oil needs to be reduced, and for this purpose, transesterification is the most widely used

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and investigated method. During the esterification process, the presence of a catalyst is mandatory for the production of biodiesel which can be used as a clean fuel in diesel engines (Hou et al., 2016). Among the catalysts, CaO shows promise, and many studies have been conducted using CaO-catalyzed transesterification (Kawashima et al., 2009; Kouzu et al., 2006; Liu et al., 2008). Seashell consist mainly of CaCO₃ (95-99.9%) with a small amount of organic matrix, and represent material that could be decomposed to CaO under high temperatures (700-1000°C) and used as catalysts for the production of biodiesel (Boey et al., 2011; Marinković et al., 2016; Salvi & Panwar, 2012; Yao et al., 2014). Perea et al. (2016) utilized waste mussel, clam and oyster shells as heterogeneous catalysts for the transesterification of Camelina sativa oil as a feedstock, into biodiesel. Biodiesel has gained considerable attention because of the potential depletion of fossil fuel resources and migrating pollutant emissions. Hence, the utilization of calcium sources from seashell waste as an economic catalyst has become a new global trend.

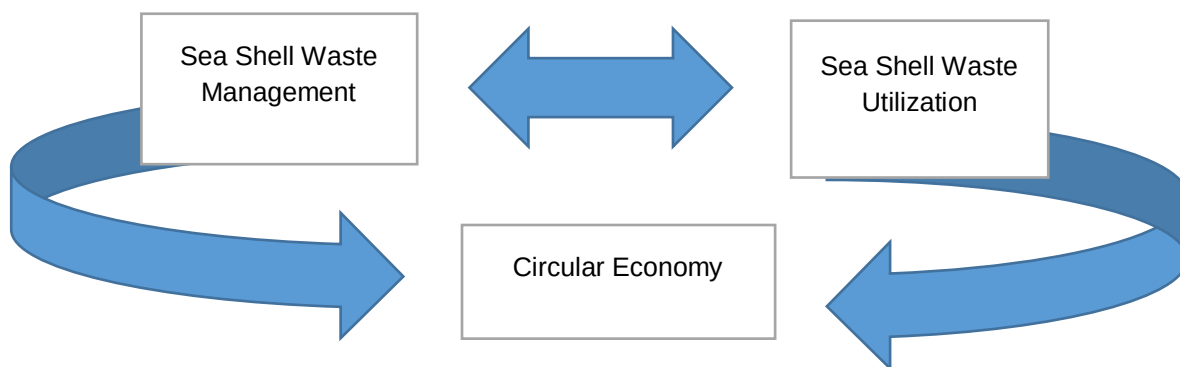
Seashells have been used in construction in many coastal settlements around the globe. The concrete made by heat-treated shells mixed with sand, water, ash, and broken shells is known as Tabby (Morris, 2017). Many authors explored the potential utilization of seashell waste in concrete in terms of re-using the discarded seashells as a replacement for conventional materials, such as cement, sand and coarse aggregate fraction (Lertwattanakruek et al., 2012; Mo et al., 2018; Olivia et al., 2015, 2017; Yoon et al., 2004). The seashell characteristics suggest that, similar to limestone, this waste could be used as an inert material in concrete (Mo et al., 2018). Investigations have shown that seashell waste can be used as a replacement for both cement and aggregate. Heating of shells at high temperature and crushing to achieve appropriate fineness are desirable treatments that improve the quality of the products (Mo et al., 2018). Despite the reduction in the workability, compressive and tensile strength and elastic modulus, it is suggested that seashell waste could still be utilized as an aggregate at a partial replacement level of up to 20% for adequate workability and strength of concrete for non-structural purposes (Mo et al., 2018; Yang et al., 2005, 2010).

One of the most interesting applications of seashells today involves shells returning to the marine environment by using them as a material for creating the artificial reef structure. Biomimicry, i.e., imitation of design and conceptual solution that can be found in nature, was the starting point and guideline for this potential application. Namely, in the natural systems, adult seashells provide a suitable substrate for larvae settling and fixing and thus forming reefs along the coast. Except as a suitable substrate for the mussels and oysters reproduction, these artificial reefs would provide shelter, an appropriate place for development and food source for many other marine organisms. They provide other ecosystem services as they create a habitat in coastal environments on which complex food webs are based and also are increasingly constructed for shoreline protection and erosion control (Wallis et al., 2016). Mussels and oysters form dense three-dimensional reef structures which can alter water flow and reduce wave action while trapping and stabilizing sediment (Borsje et al., 2011; Wallis et al., 2015). Wallis et al. (2016) showed that the persistence and sustainability of artificially constructed oyster reefs are strongly

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dependent on the local environmental conditions. They concluded that the artificial reef structure provides a substrate which facilitates reef establishment, but that tidal emersion is an essential factor that can be used to predict where artificial oyster reefs have the potential to develop in self-sustaining reefs and to contribute to coastal protection.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)



13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Locale of the study: This study will be conducted in Roxas City, Capiz more specifically in its 23 coastal barangays: Barangay 7, baliuagan, Bato, Banica, Barra, Baybay, Cagay, Cogon, Culajao, Culasi, Dayao, Dinginan, Dumolog, Gabuan, Libas, Mongpong, Olotaya, Punt Cogon, Punta Tabuc, Talon, Sibaguan, Tanque, and Tanza. Roxas City is chosen because of its abundant shellfish produced.

Research Design and Respondents: This study will employ descriptive research design and will utilize both primary data and secondary data. Respondents of this study will be the identified shell industry: shellmeat processors, shellfish vendors, and other industry.

Data Collection and Analysis. The Primary data will be gathered from interview and field observation as to seashell waste management and practices. Secondary data will be gathered from City Agriculture Office and LGU. Frequency count and percentage will be used to analyze and present the data.

The activities to be employed in this study are:

1. *Preliminary survey*

Actual site visit for field observation. A letter of request will be send to different coastal barangays before the conduct of the study.

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Collection of data of number of post-harvest processors involved shell industry. Data on the number of processors involved in shell industry will be determine from the LGU.

* *post-harvest processors*- shell meat processors, vendors, restaurants, handicraft

2. *Identification of respondents.* Respondents of the study will be identified according to post-harvest processors*.
3. *Interview.* Structured interview will be conducted to determine the volume and type of sea shell waste generated, waste management, disposal, and practices, and other risks and issues.
4. *Focus Group Discussion.* FGD will be conducted to triangulate the data gathered. This will be composed of respondents of the study, Barangay Fishery and Aquaculture Resource Management Council (BFARMC), and barangay officials.
5. Data gathered will be analyzed using frequency count and percentage.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	2022					2023						
	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	July
3. Procurement of materials and tools												
4. Laboratory Work												
5. Data gathering												
6. Data Analysis												
7. Research paper writing												
8. Colloquim/Defense												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

The study is expected to produce:

1. Baseline data of shellwaste generation and management in Roxas City.
2. Development of management plan for shell waste.
3. Development of utilization plan of shell waste as input to circular economy.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

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- I. Basic Research
- II. Development of Sustainable Plan
- III. Extension Services
- IV. Public Use

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

This study may contribute to the waste management plan and valorization of sea shells waste for circular economy.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

	Counterpart Funding	Other Funding	TOTAL
I. Personal Services Salaries/ Honorarium			
II. Maintenance and Other Operating Expenses			
A. Direct Cost			
Travel Expenses (Local)			P 15, 000
Supplies and Materials			
Raw Materials			
Tools			
Facility and Equipment Rental Expenses			
Documentation Printing, Photocopying Expenses			P 10, 000
B. Indirect Cost			
Utilities/Miscellaneous			P 5, 000
GRAND TOTAL			P 30, 000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Chilakala, Ramakrishna, Chottitisupawong Thannaree, Eunsoo Justin Shin, Thriveni Thenepalli, and Ji Whan Ahn. 2019. "Sustainable Solutions for Oyster Shell Waste

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<https://doi.org/10.3390/recycling4030035>.

Hene I. Hapinat, 2019. Waste from Discarded Oyster Shells: A promising Raw Material for Lime Industry in the Philippines.

Mihajlo JOVIĆ^{1*}, Milica MANDIĆ² , Marija ŠLJIVIĆ-IVANOVIĆ¹ & Ivana SMIČIKLAS¹. Recent trends in application of shell waste from mariculture. Institute of Marine Biology, University of Montenegro, P.O. Box 69, 85330 Kotor, Montenegro.

Paceño, B. et al. 2021, Study of seashell waste recycling in fireproofing material: Technical, environmental, and economic assessment.
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Espejo-Hermes, 1985. Fish Processing Technology in the Tropics.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)